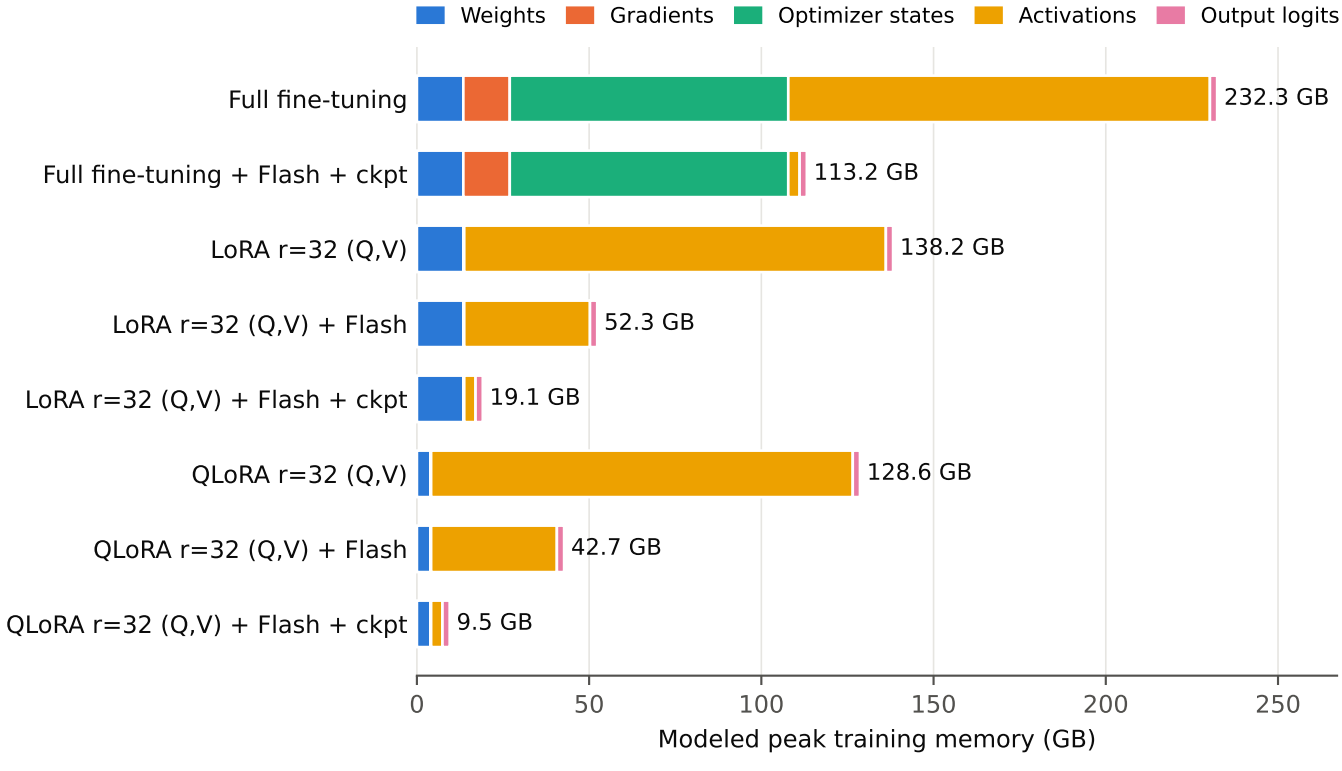
Weights
Gradients
Optimizer states
Activations
Output logits
Full fine-tuning
232.3 GB
Full fine-tuning + Flash + ckpt
113.2 GB
LoRA r=32 (Q,V)
138.2 GB
LoRA r=32 (Q,V) + Flash
52.3 GB
LoRA r=32 (Q,V) + Flash + ckpt
19.1 GB
QLoRA r=32 (Q,V)
128.6 GB
QLoRA r=32 (Q,V) + Flash
42.7 GB
QLoRA r=32 (Q,V) + Flash + ckpt
9.5 GB
0
50
100
150
200
250
Modeled peak training memory (GB)